THE LANCET

Supplementary appendix

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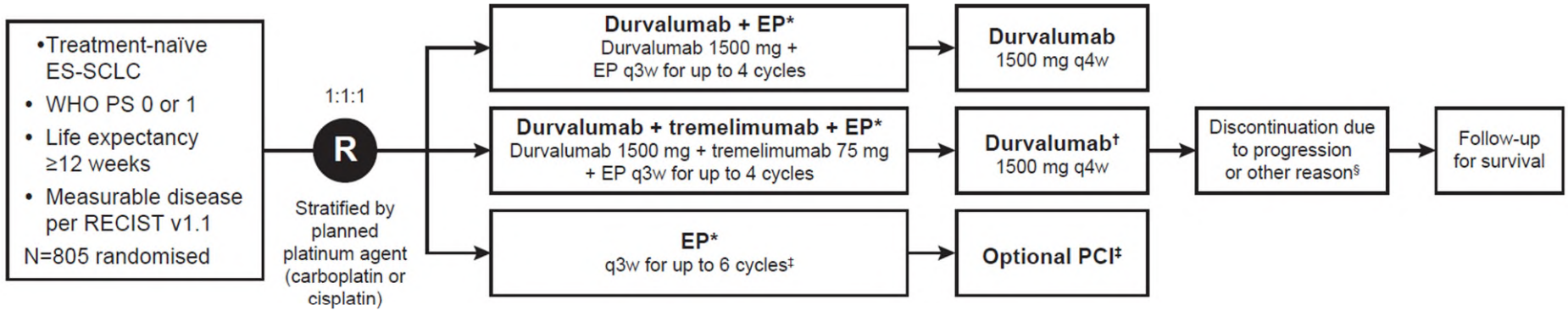
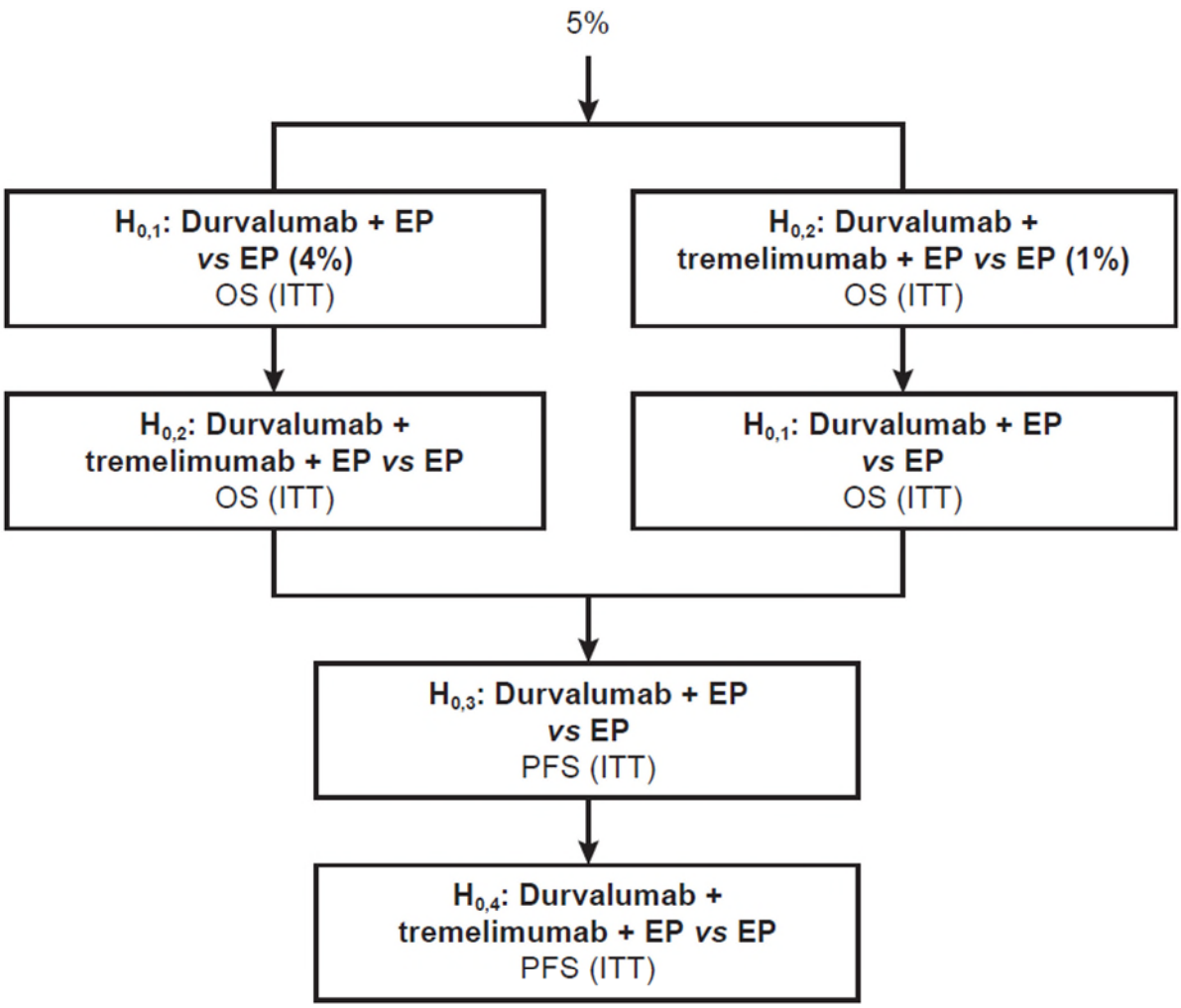
Figure S1: **CASPIAN study design**



*EP consists of etoposide 80–100 mg/m^2 with either carboplatin AUC 5–6 or cisplatin 75–80 mg/m^2. †Patients received an additional dose of tremelimumab post-EP. ‡Patients could receive an additional 2 cycles of EP (up to 6 cycles total) in the control arm, as well as PCI post-EP at the investigator's discretion. §Patients continued treatment until disease progression per investigator assessment, unacceptable toxicity, evidence of a new paraneoplastic syndrome or worsening of an existing paraneoplastic syndrome, pregnancy or intent to become pregnant, initiation of alternative anticancer therapy including another investigational agent, noncompliance, or withdrawal of consent. Patients in all arms who had disease progression per RECIST v1.1 (unconfirmed and confirmed) who, in the investigator's opinion, would continue to receive benefit from their assigned treatment and who met the criteria for treatment in the setting of progressive disease could continue to receive their assigned treatment for as long as they were judged to be gaining clinical benefit. This included EP; however, EP was restricted to a maximum of 4 cycles for patients in the immunotherapy arms and a maximum of 6 cycles for patients in the control arm.

AUC=area under the curve; ES-SCLC=extensive-stage small-cell lung cancer; EP=platinum-etoposide; PCI=prophylactic cranial irradiation; PS=performance status; q3w=every 3 weeks; q4w=every 4 weeks; RECIST v1.1=Response Evaluation Criteria in Solid Tumours version 1.1.

Figure S2: **Hierarchical multiple testing procedure**



A hierarchical multiple testing procedure with a gatekeeping strategy was used to control the type I error at a two-sided 5% significance level. The hypotheses were to be tested using a multiple testing procedure with an alpha-exhaustive recycling strategy.[1] This strategy was used to test the two primary analyses of OS and two secondary analyses of PFS. PFS was therefore only to be tested within the multiple testing procedure if both OS primary analyses achieved significance. The overall 5% alpha was split between the primary endpoints: an alpha level of 4% was allocated to the analysis of OS for durvalumab plus EP versus EP, and an alpha level of 1% for the analysis of OS for durvalumab plus tremelimumab plus EP versus EP.

EP=platinum-etoposide; H=hypothesis; ITT=intention-to-treat; OS=overall survival; PFS=progression-free survival.

Table S1: **Inclusion and exclusion criteria**

Inclusion criteria
Male or female ≥18 years at the time of screening. In Japan, patients must be aged ≥20 years at the time of screening.
Written informed consent and any locally required authorisation (eg, Health Insurance Portability and Accountability Act in the USA, European Union [EU] Data Privacy Directive in the EU) obtained from the patient/legal representative prior to performing any protocol-related procedures, including screening evaluations.
Histologically or cytologically documented ES-SCLC (American Joint Committee on Cancer [7th edition][2] stage IV [T any, N any, M1 a/b], or T3–4 due to multiple lung nodules that are too extensive or have tumour/nodal volume that is too large to be encompassed in a tolerable radiation plan). Brain metastases must be asymptomatic or treated and stable off steroids and anti-convulsants for at least 1 month prior to study treatment. Patients with suspected brain metastases at screening should have a CT/MRI of the brain prior to study entry.
Provision of an archived tumour tissue block (or at least 15 newly cut unstained slides) where such samples exist.
Patients must be considered suitable to receive a platinum-based chemotherapy regimen as first-line treatment for ES-SCLC. Chemotherapy must contain either carboplatin or cisplatin in combination with etoposide.
WHO performance-status score of 0 or 1 at enrollment.
Life expectancy ≥12 weeks at randomisation (day 1).
Body weight >30 kg.
At least 1 lesion, not previously irradiated, that can be accurately measured at baseline as ≥10 mm in the longest diameter (except lymph nodes, which must have a short axis ≥15 mm) with CT or MRI and that is suitable for accurate repeated measurements as per Response Evaluation Criteria in Solid Tumours version 1.1 guidelines.
No prior exposure to immune-mediated therapy including, but not limited to, other anti-cytotoxic T-lymphocyte-associated antigen-4, anti-programmed cell death-1, anti-programmed cell death ligand-1, and anti-programmed cell death ligand-2 antibodies, excluding therapeutic anticancer vaccines.

Adequate organ and marrow function as defined below:

- Haemoglobin ≥9·0 g/dL.
- Absolute neutrophil count ≥1·5 × 10^9/L (use of granulocyte colony-stimulating factor is not permitted at screening).
- Platelet count ≥100 × 10^9/L.
- Serum bilirubin ≤1·5 × the ULN. This will not apply to patients with confirmed Gilbert's syndrome, who will be allowed in consultation with their physician.
- In patients without hepatic metastasis: ALT and AST ≤2·5 × ULN; for patients with hepatic metastases, ALT and AST ≤5 × ULN.
- Measured or calculated creatinine clearance: >60 mL/min for patients on cisplatin and >45 mL/min for patients on carboplatin, as determined by Cockcroft–Gault (using actual body weight).

Males:

$$\text{Creatinine clearance (mL/min)} = \frac{\text{Weight (kg)} \times (140 - \text{Age})}{72 \times \text{serum creatinine (mg/dL)}}$$

Females:

$$\text{Creatinine clearance (mL/min)} = \frac{\text{Weight (kg)} \times (140 - \text{Age})}{72 \times \text{serum creatinine (mg/dL)}} \times 0{\cdot}85$$

Evidence of post-menopausal status or negative urinary or serum pregnancy test for female pre-menopausal patients. Women will be considered post-menopausal if they have been amenorrhoeic for 12 months without an alternative medical cause. The following age-specific requirements apply:

- Women <50 years of age would be considered post-menopausal if they have been amenorrhoeic for 12 months or more following cessation of exogenous hormonal treatments and if they have luteinising hormone and follicle stimulating hormone levels in the post-menopausal range for the institution or underwent surgical sterilisation (bilateral oophorectomy or hysterectomy).
- Women ≥50 years of age would be considered post-menopausal if they have been amenorrhoeic for 12 months or more following cessation of all exogenous hormonal treatments, had radiation-induced menopause with last menses >1 year ago, had chemotherapy-induced menopause with last menses >1 year ago, or underwent surgical sterilisation (bilateral oophorectomy, bilateral salpingectomy, or hysterectomy).

Exclusion criteria
Involvement in the planning and/or conduct of the study (applies to both AstraZeneca staff and/or staff at the study site).
Previous investigational product assignment in the present study.
Concurrent enrollment in another clinical study, unless it is an observational (non-interventional) clinical study or the follow-up period of an interventional study.
Medical contraindication to platinum-etoposide.
Participation in another clinical study with an investigational product during the last 4 weeks.
Any concurrent chemotherapy, investigational product, biologic, or hormonal therapy for cancer treatment. Concurrent use of hormonal therapy for non-cancer-related conditions (eg, hormone replacement therapy) is acceptable.
Any history of radiotherapy to the chest prior to systemic therapy or planned consolidation chest radiation therapy. Radiation therapy outside of the chest for palliative care (ie, bone metastasis) is allowed but must be completed before first dose of the study medication.
Major surgical procedure (as defined by the investigator) within 28 days prior to the first dose of investigational product. Local surgery of isolated lesions for palliative intent is acceptable.
History of allogenic organ transplantation.
Has a PNS of autoimmune nature, requiring systemic treatment (systemic steroids or immunosuppressive agents) or has a clinical symptomatology suggesting worsening of PNS.
Active or prior documented autoimmune or inflammatory disorders (including inflammatory bowel disease [eg, colitis or Crohn's disease], diverticulitis with the exception of diverticulosis, systemic lupus erythematosus, sarcoidosis syndrome, or Wegener syndrome (granulomatosis with polyangiitis, Graves' disease, rheumatoid arthritis, hypophysitis, and uveitis, etc.). The following are exceptions to this criterion: • Patients with vitiligo or alopecia. • Patients with hypothyroidism (eg, following Hashimoto syndrome) stable on hormone replacement • Any chronic skin condition that does not require systemic therapy. • Patients without active disease in the last 5 years may be included but only after consultation with the study physician. • Patients with celiac disease controlled by diet alone.

Uncontrolled intercurrent illness, including but not limited to, ongoing or active infection, interstitial lung disease, symptomatic congestive heart failure, uncontrolled hypertension, unstable angina pectoris, cardiac arrhythmia, serious chronic gastrointestinal conditions associated with diarrhoea, or psychiatric illness/social situations that would limit compliance with study requirement, substantially increase risk of incurring adverse events or compromise the ability of the patient to give written informed consent.
History of another primary malignancy except for: • Malignancy treated with curative intent and with no known active disease ≥5 years before the first dose of the investigational product and of low potential risk for recurrence. • Adequately treated nonmelanoma skin cancer or lentigo maligna without evidence of disease. • Adequately treated carcinoma in situ without evidence of disease (eg, cervical cancer in situ).
History of leptomeningeal carcinomatosis.
History of active primary immunodeficiency.
Active infection, including tuberculosis (clinical evaluation that includes clinical history, physical examination and radiographic findings, and tuberculosis testing in line with local practice), HBV (known positive HBV surface antigen result), HCV, or HIV (positive HIV 1/2 antibodies). Patients with a past or resolved HBV infection (defined as the presence of HBV core antibody and absence of HBV surface antigen) are eligible. Patients positive for HCV antibody are eligible only if polymerase chain reaction is negative for HCV ribonucleic acid.
Current or prior use of immunosuppressive medication within 14 days before the first dose of durvalumab or tremelimumab. The following are exceptions to this criterion: • Intranasal, inhaled, topical steroids, or local steroid injections (eg, intra-articular injection). • Systemic corticosteroids at physiologic doses not to exceed 10 mg/day of prednisone or its equivalent. • Steroids as premedication for hypersensitivity reactions (eg, CT scan premedication). Premedication with steroids for chemotherapy is acceptable.
Receipt of live, attenuated vaccine within 30 days prior to the first dose of investigational product. Patients, if enrolled, should not receive live vaccine whilst receiving the investigational product and up to 30 days after the last dose of investigational product.

Female patients who are pregnant or breast-feeding or male or female patients of reproductive potential who are not willing to employ effective birth control from screening to 90 days after the last dose of durvalumab monotherapy or 180 days after the last dose of durvalumab plus tremelimumab combination therapy.
Known allergy or hypersensitivity to durvalumab, tremelimumab, etoposide, carboplatin, cisplatin, or any of their excipients.
Prior randomisation or treatment in a previous durvalumab and/or tremelimumab clinical study regardless of treatment arm assignment.

ALT=alanine aminotransferase; AST=aspartate aminotransferase; CT=computed tomography; ES-SCLC=extensive-stage small-cell lung cancer; HBV=hepatitis B virus; HCV=hepatitis C virus; HIV=human immunodeficiency virus; MRI=magnetic resonance imaging; PNS=paraneoplastic syndrome; ULN=upper limit of normal.

Table S2: **Sample size assumptions**

Primary endpoint	**Power (%)**	**HR**	**Events**	**Overall 2-sided significance level (%)**
OS (durvalumab plus EP *vs* EP)	96	0·69	425	4*
OS (durvalumab plus tremelimumab plus EP *vs* EP)	89	0·69	425	1*

*Adjusting for one interim analysis of OS planned when 75% of target OS events had occurred and final analysis of OS, using the Lan–DeMets spending function that approximates an O'Brien−Fleming approach to account for multiple comparisons.[3]

EP=platinum-etoposide; HR=hazard ratio; OS=overall survival.

Table S3: **Important protocol deviations in the intention-to-treat population**

Important protocol deviation	**Durvalumab + EP (n=268)**	**EP (n=269)**
≥1 important deviation, n	11	8
Baseline imaging was performed more than 28 days before start of study treatment	0	1
Baseline tumor assessments (RECIST v1.1) performed more than 42 days before the date of randomisation	0	1
Patients randomised but who did not receive study treatment	3	3
Patients who deviated from key entry criteria as per the protocol	6	3
Received prohibited concomitant medications (including other anticancer agents)	1	1
Other important deviation	1*	0

Important protocol deviations were those that could substantially affect the completeness, accuracy, and/or reliability of the study data, or a patient's rights, safety, or wellbeing. Note that the same patient may have had more than one important protocol deviation. *Tremelimumab was prepared for a patient on a non-tremelimumab arm but was never administered to the patient.

EP=platinum-etoposide; RECIST v1.1=Response Evaluation Criteria in Solid Tumours version 1.1.

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Table S4: **Subsequent anticancer therapy**

	Durvalumab + EP (n=268)	**EP (n=269)**
Patients who received study treatment, n (%)	265 (99%)	266 (99%)
Patients ongoing study treatment	43 (16%)	0
Patients receiving any subsequent therapy, n (%)	113 (42%)	119 (44%)
Chemotherapy	110 (41%)	112 (42%)
Single agent	55 (21%)	65 (24%)
Platinum doublet	51 (19%)	43 (16%)
Other	22 (8%)	30 (11%)
Immunotherapy	5 (2%)	14 (5%)
Single agent	0	4 (1%)
Immunotherapy + immunotherapy	2 (1%)	3 (1%)
Immunotherapy + chemotherapy	0	1 (<1%)
Investigational agent	3 (1%)	6 (2%)
Other systemic therapies	1 (<1%)	5 (2%)
Patients receiving 1 subsequent line of treatment, n (%)	113 (42%)	119 (44%)
Patients receiving 2 subsequent lines of treatment, n (%)	33 (12%)	39 (14%)
Patients receiving >2 subsequent lines of treatment, n (%)	6 (2%)	8 (3%)

Data cutoff date: March 11, 2019.

EP=platinum-etoposide.

Table S5: **Treatment-related adverse events as assessed per the investigator (safety population)**

	Durvalumab + EP (n=265)		**EP (n=266)**	
	Any grade	**Grade 3 or 4**	**Any grade**	**Grade 3 or 4**
Any treatment-related event, n (%)	237 (89%)	121 (46%)	240 (90%)	138 (52%)
Any treatment-related serious event, n (%)	35 (13%)	25 (9%)	50 (19%)	45 (17%)
Any treatment-related event leading to discontinuation, n (%)*	15 (6%)	3 (1%)	13 (5%)	4 (2%)
Any treatment-related event leading to death, n (%)†	5 (2%)	–	2 (1%)	–
Treatment-related adverse events with an incidence of ≥5% in any grade category or events of grade 3 or 4 with an incidence of ≥2% in either arm, n (%)‡				
Neutropenia	104 (39%)	61 (23%)	116 (44%)	86 (32%)
Anaemia	85 (32%)	21 (8%)	103 (39%)	38 (14%)
Alopecia	74 (28%)	3 (1%)	85 (32%)	2 (1%)
Nausea	74 (28%)	0	75 (28%)	5 (2%)
Thrombocytopenia	37 (14%)	14 (5%)	48 (18%)	24 (9%)
Vomiting	32 (12%)	0	38 (14%)	2 (1%)
Decreased appetite	33 (12%)	1 (<1%)	35 (13%)	1 (<1%)
Leucopenia	36 (14%)	15 (6%)	32 (12%)	14 (5%)
Fatigue	30 (11%)	2 (1%)	36 (14%)	3 (1%)
Asthenia	27 (10%)	2 (1%)	31 (12%)	2 (1%)

	Durvalumab + EP (n=265)		EP (n=266)	
	Any grade	**Grade 3 or 4**	**Any grade**	**Grade 3 or 4**
Neutrophil count decreased	24 (9%)	16 (6%)	28 (11%)	17 (6%)
Constipation	23 (9%)	0	24 (9%)	0
Diarrhoea	17 (6%)	2 (1%)	15 (6%)	2 (1%)
Febrile neutropenia	15 (6%)	13 (5%)	17 (6%)	17 (6%)
White blood cell count decreased	12 (5%)	4 (2%)	17 (6%)	6 (2%)
Platelet count decreased	14 (5%)	4 (2%)	13 (5%)	6 (2%)
Paraesthesia	14 (5%)	0	11 (4%)	0
Hypothyroidism	23 (9%)	0	0	0
Hyperthyroidism	22 (8%)	0	0	0
Lipase increased	10 (4%)	8 (3%)	2 (1%)	1 (<1%)

Data cutoff date: March 11, 2019. Listed are all adverse events assessed by the investigator as possibly related to any study treatment that occurred during the treatment period and up to 90 days after the last dose of durvalumab or EP or up to the start of any subsequent therapy (whichever occurred first). *Includes patients who permanently discontinued at least one study drug. [†]Treatment-related adverse events leading to death in the durvalumab plus EP arm were cardiac arrest, dehydration, hepatotoxicity, pancytopenia, and sepsis in one patient each. Treatment-related adverse events leading to death in the EP arm were pancytopenia and thrombocytopenia/haemorrhage in one patient each. [‡]The events are listed in descending order of frequency across both treatment arms.

EP=platinum-etoposide.

Table S6: **Serious adverse events occurring in ≥2 patients in any treatment arm (safety population)**

	Durvalumab + EP (n=265)	**EP (n=266)**
Any serious event, n (%)*	82 (31%)	96 (36%)
Febrile neutropenia	12 (5%)	12 (5%)
Anaemia	5 (2%)	12 (5%)
Pneumonia	6 (2%)	9 (3%)
Thrombocytopenia	1 (<1%)	9 (3%)
Neutropenia	2 (1%)	7 (3%)
Pancytopenia	4 (2%)	3 (1%)
Hyponatraemia	2 (1%)	4 (2%)
Pneumonitis	3 (1%)	3 (1%)
Diarrhoea	1 (<1%)	4 (2%)
Acute kidney injury	2 (1%)	2 (1%)
Chronic obstructive pulmonary disease	3 (1%)	1 (<1%)
Pleural effusion	2 (1%)	2 (1%)
Cerebrovascular accident	0	3 (1%)
General physical health deterioration	2 (1%)	1 (<1%)
Hypokalaemia	0	3 (1%)
Respiratory tract infection	2 (1%)	1 (<1%)
Sepsis	2 (1%)	1 (<1%)
Sudden death	2 (1%)	1 (<1%)
Syncope	1 (<1%)	2 (1%)
Transient ischaemic attack	2 (1%)	1 (<1%)
Upper respiratory tract infection	2 (1%)	1 (<1%)
Vomiting	0	3 (1%)
Acute myocardial infarction	0	2 (1%)
Atrial fibrillation	2 (1%)	0
Constipation	2 (1%)	0

	Durvalumab + EP (n=265)	**EP (n=266)**
Death	0	2 (1%)
Deep vein thrombosis	2 (1%)	0
Dyspnoea	0	2 (1%)
Lung infection	0	2 (1%)
Nausea	0	2 (1%)
Pyrexia	0	2 (1%)
Septic shock	2 (1%)	0
Type 1 diabetes mellitus	2 (1%)	0

Data cutoff date: March 11, 2019. Listed are all serious adverse events that occurred during the treatment period and up to 90 days after the last dose of durvalumab or EP or up to the start of any subsequent therapy (whichever occurred first). *The events are listed in descending order of frequency across both treatment arms. EP=platinum-etoposide.

Table S7: **Adverse events leading to treatment discontinuation (safety population)**

	Durvalumab + EP (n=265)	**EP (n=266)**
Any event leading to discontinuation, n (%)*,†	25 (9%)	25 (9%)
Acute kidney injury	3 (1%)	4 (2%)
Neutropenia	1 (<1%)	2 (1%)
Sudden death	2 (1%)	1 (<1%)
Thrombocytopenia	0	3 (1%)
Cardiac arrest	1 (<1%)	1 (<1%)
Deafness	0	2 (1%)
Pancytopenia	1 (<1%)	1 (<1%)
Acute cardiac failure	0	1 (<1%)
Acute respiratory failure	0	1 (<1%)
Anaemia	1 (<1%)	0
Aspiration	1 (<1%)	0
Asthenia	1 (<1%)	0
Atrial fibrillation	0	1 (<1%)
Bacterial urinary tract infection	1 (<1%)	0
Cardiopulmonary failure	0	1 (<1%)
Cerebrovascular accident	0	1 (<1%)
Death	0	1 (<1%)
Diabetic ketoacidosis	1 (<1%)	0
Diarrhoea	1 (<1%)	0
Haematotoxicity	0	1 (<1%)
Haemorrhage	0	1 (<1%)
Hepatic failure	1 (<1%)	0
Hepatitis C	1 (<1%)	0
Hepatotoxicity	1 (<1%)	0
Infectious pleural effusion	0	1 (<1%)

	Durvalumab + EP (n=265)	**EP (n=266)**
Ischaemic stroke	0	1 (<1%)
Lung disorder	1 (<1%)	0
Multiple organ dysfunction syndrome	0	1 (<1%)
Muscular weakness	1 (<1%)	0
Nephropathy	1 (<1%)	0
Neurotoxicity	1 (<1%)	0
Pneumonia	0	1 (<1%)
Pneumonitis	1 (<1%)	0
Pulmonary artery thrombosis	1 (<1%)	0
Pulmonary embolism	1 (<1%)	0
Renal failure	0	1 (<1%)
Seizure	0	1 (<1%)
Sepsis	1 (<1%)	0
Sudden cardiac death	0	1 (<1%)
Tinnitus	1 (<1%)	0
Tumour lysis syndrome	0	1 (<1%)

Data cutoff date: March 11, 2019. Listed are all adverse events leading to discontinuation that occurred during the treatment period and up to 90 days after the last dose of durvalumab or EP or up to the start of any subsequent therapy (whichever occurred first). *Includes patients who permanently discontinued at least one study drug. [†]The events are listed in descending order of frequency across both treatment arms. EP=platinum-etoposide.

Table S8: **Immune-mediated adverse events (grouped terms) (safety population)**

	Durvalumab + EP (n=265)		**EP (n=266)**	
	Any grade*	**Grade 3 or 4**	**Any grade***	**Grade 3 or 4**
Any immune-mediated adverse event (grouped term), n (%)[†,‡]	52 (20%)	12 (5%)	7 (3%)	1 (<1%)
Hypothyroid events	24 (9%)	0	2 (1%)	0
Hyperthyroid events	14 (5%)	0	0	0
Pneumonitis	7 (3%)	2 (1%)	2 (1%)	1 (<1%)
Hepatic events	7 (3%)	5 (2%)	0	0
Dermatitis/rash	4 (2%)	0	2 (1%)	0
Diarrhoea/colitis	4 (2%)	1 (<1%)	1 (<1%)	0
Thyroiditis	4 (2%)	0	0	0
Type 1 diabetes mellitus	4 (2%)	4 (2%)	0	0
Adrenal insufficiency	1 (<1%)	0	0	0
Pancreatic events	1 (<1%)	1 (<1%)	0	0
Other rare/miscellaneous[§]	2 (1%)	0	0	0

Data cutoff date: 11 March 2019. Listed are all immune-mediated adverse events that occurred during the treatment period and up to 90 days after the last dose of durvalumab or EP or up to the start of any subsequent therapy (whichever occurred first). *Grade 5 immune-mediated adverse events occurred in one patient receiving durvalumab plus EP (hepatotoxicity) and one patient receiving EP (pneumonitis). [†]An immune-mediated adverse event is defined as an event that is associated with drug exposure and

consistent with an immune-mediated mechanism of action, where there is no clear alternate aetiology and the event required treatment with systemic corticosteroids or other immunosuppressants and/or, for specific endocrine events, endocrine therapy. [‡]The events are listed in descending order of frequency across both treatment arms. [§]The events in the 'other rare/miscellaneous' category were both arthritis.

EP=platinum-etoposide.

Supplementary references

1 Burman CF, Sonesson C, Guilbaud O. A recycling framework for the construction of Bonferroni-based multiple tests. *Stat Med* 2009; **28:** 739–61.

2 Edge SB, Byrd DR, Compton CC, Fritz AG, Greene FL, Trotti A (eds). AJCC cancer staging manual (7th ed). New York, NY: Springer; 2010.

3 Lan K-KG, DeMets DL. Discrete sequential boundaries for clinical trials. *Biometrika* 1983; **70:** 659–63.